

## Chapter 2, Problem 5RQ

### Problem

A network has 12 branches and 8 independent loops. How many nodes are there in the network?

- (a) 19
- (b) 17
- (c) 5
- (d) 4

### Step-by-step solution

#### Step 1 of 2

A node is the point of connection between two or more branches.

A loop is any closed path in the circuit.

Recall the fundamental theorem of network topology.

A network with  $b$  branches,  $n$  nodes and  $l$  independent loops satisfies the equation.

$$b = l + n - 1 \dots\dots (1)$$

#### Step 2 of 2

The network has 12 branches and 8 independent loops.

Determine the nodes in the network.

Substitute 12 for  $b$ , 8 for  $l$  in equation (1).

$$b = l + n - 1$$

$$12 = 8 + n - 1$$

$$n = 5$$

Therefore, the nodes in the network is 5.

Therefore, option (c) is correct.